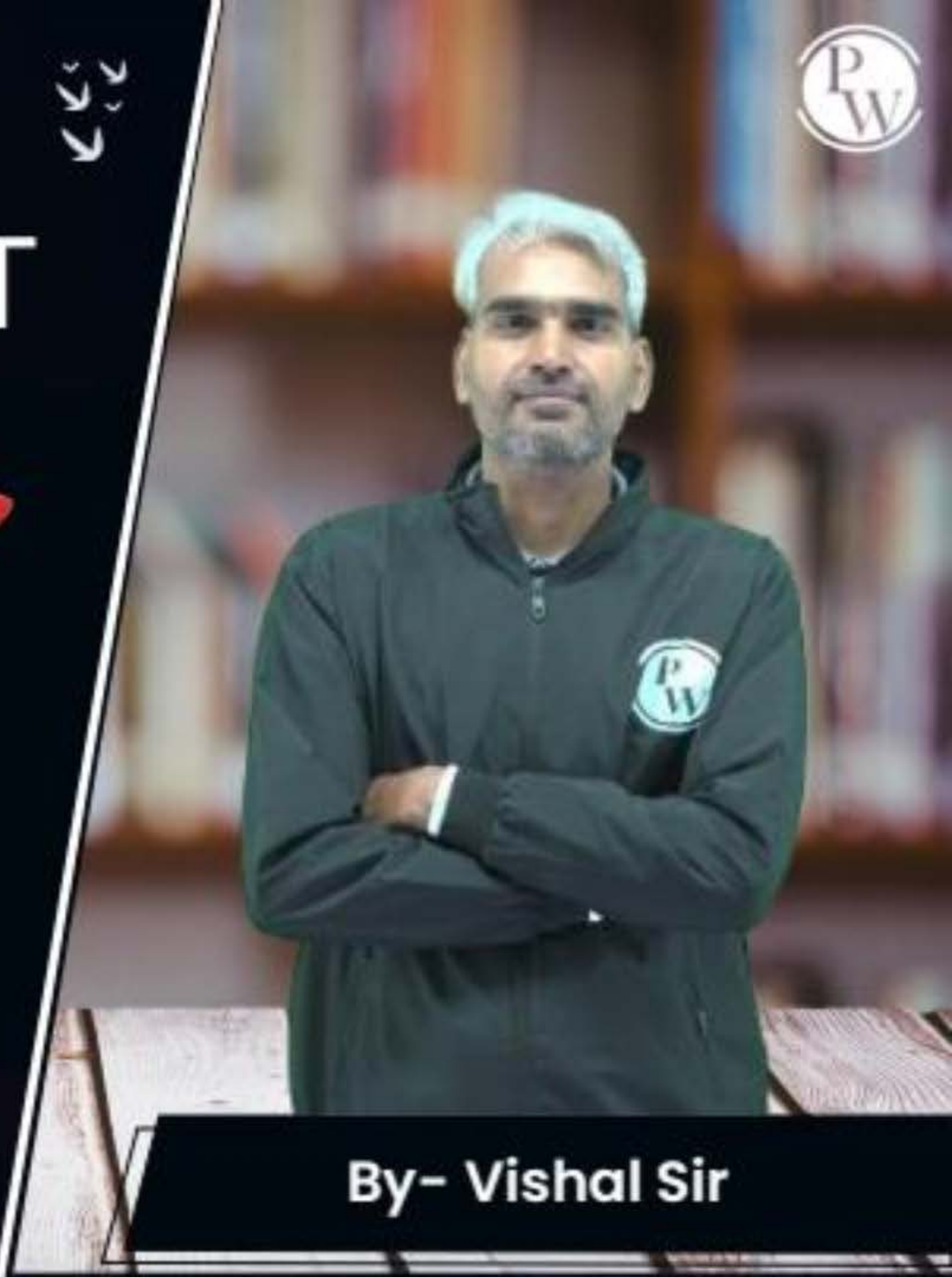
Computer Science & IT



Database Management System

Relational Model & Normal Forms

Lecture No. 09



By- Vishal Sir

Recap of Previous Lecture



- Topic: Membership test
- Topic: Relationship between two FD sets
- Topic: FD set of a sub-relation

Slide

Topics to be Covered



Topic Minimal cover or Canonical cover

Slide

H.W.

#Q. Consider a relational schema R(A,B,C,D,E,F) with FD set

☆ F = {AB→C, B→D, BC→A, D→EF}

Find the FD set F1 for sub-relation R1(A,B,C,D) of R(A,B,C,D,E,F).

Also find candidate keys for the sub-relation R1(A,B,C,D).

$F_1 = \{B \to D,\ AB \to CD,\ BC \to AD,\ ABC \to D,\ ABD \to C,\ BCD \to A\}$

C.Ks of R_1 are (AB) & (BC)

$(A)^+ = \{A\}$

$(B)^+ = \{B, D, E, F\}$ B→D

$(C)^+ = \{C\}$

$(D)^+ = \{D, E, F\}$

$(AB)^+ = \{A, B, C, D, E, F\}$ AB→CD

$(AC)^+ = \{A, C\}$

$(AD)^+ = \{A, D, E, F\}$

$(BC)^+ = \{B, C, A, D, E, F\}$ BC→AD

$(BD)^+ = \{B, D, E, F\}$

$(CD)^+ = \{C, D, E, F\}$

$(ABC)^+ = \{A, B, C, D, E, F\}$ ABC→D

$(ABD)^+ = \{A, B, C, D, E, F\}$ ABD→C

$(ACD)^+ = \{A, C, D, E, F\}$

$(BCD)^+ = \{B, C, A, D, E, F\}$ BCD→A



Topic : Minimal cover (Canonical cover)

- Minimal cover of canonical cover of FD set F is a set of functional dependencies (F_m)
  such that,
  - $F_m = F$ and
  - F_m does not contain any redundant FD, and F_m must not contain any extraneous attribute at either side of any of its FD

We need no check only w.r.t. LHS. if we have already simplified FDs of FD set F.



Topic : Procedure to obtain minimal cover of FD set

1. Simplify RHS of all FDs (i.e., split the FDs such that RHS contain exactly one attribute)
2. For all FDs find redundant (extraneous) attribute in LHS.
3. Eliminate all redundant FDs
4. Apply Union if needed
5. The result is minimal Cover

As soon as any redundant FD is identified, remove it from the set

As soon as any extraneous attribute is identified, remove that attribute and Check for other extraneous attributes



Topic : Testing if an Attribute is Extraneous in LHS

Set of attributes

- Consider a set F of functional dependencies and functional dependency $\alpha \rightarrow \beta$ in F.

set of attributes

- To test if attribute $A \in \alpha$ is extraneous in α (i.e., Any extraneous attribute in LHS of FD)
  1. compute $(\{\alpha\} - \{A\})^+$ using the dependencies in F', where $F' = (F - \{\alpha \rightarrow \beta\})$
  2. check if $(\{\alpha\} - A)^+$ contains A; if it does then, A is extraneous.

FDs except $\alpha \rightarrow \beta$

& , if not then A is not extraneous

in $\alpha \rightarrow \beta$, if $|\alpha| = 1$, then no extraneous attribute in LHS.

#e.g., Consider the FD set

$$F = \{AB \rightarrow C,\ B \rightarrow A,\ A \rightarrow B\}$$

$$F: \begin{Bmatrix} AB \rightarrow C \\ B \rightarrow A \\ A \rightarrow B \end{Bmatrix} \xrightarrow[\text{Simplify R.Hs}]{\text{Step-1,}} \begin{Bmatrix} AB \rightarrow C \\ B \rightarrow A \\ A \rightarrow B \end{Bmatrix}$$

Eliminate Extraneous attribute from LHS

Check if 'A' is Extraneous in $\underset{\alpha}{AB} \rightarrow \underset{\beta}{C}$

$\alpha - \{A\} = \{A,B\} - \{A\} = B$

$(\alpha - \{A\})^+ = (B)^+$ w.r.t. $(F - (AB \rightarrow C)) = \{B, A\}$

$A \in (B)^+$

∴ A is extraneous in $AB \rightarrow C$, if B is present

Eliminate Extraneous attribute from LHS

Check if 'B' is extraneous in $AB \rightarrow C$

$(A)^+$ w.r.t. $(F - (AB \rightarrow C)) = \{A, B\}$

$B \in (A)^+$

∴ B is extraneous if A is present

∴ Set of FDs after removal of extraneous attribute from L.H.S $= \begin{Bmatrix} B \rightarrow C \\ B \rightarrow A \\ A \rightarrow B \end{Bmatrix}$

∴ Set of functional dependencies after removal of Extraneous attributes from L.H.S $= \begin{Bmatrix} A \rightarrow C \\ B \rightarrow A \\ A \rightarrow B \end{Bmatrix}$



Topic : Testing if any FD is redundant

Set of attributes

- Consider a set F of functional dependencies and functional dependency $\alpha \rightarrow \beta$ in F.

∴ Consider 'β' is single attribute" Because we have already simplified w.r.t R.H.S.

- To test if $\alpha \rightarrow \beta$ is redundant
  1. compute α^+ using only the dependencies in F', where F' = (F – {$\alpha \rightarrow \beta$})
  2. check that α^+ contains β; if it does then, $\alpha \rightarrow \beta$ is redundant

if not, then $\alpha \rightarrow \beta$ is not redundant.

#e.g., Consider the FD set

$$F = \{AB \rightarrow C,\ B \rightarrow A,\ A \rightarrow B\}$$

$$F: \begin{cases} AB \rightarrow C \\ B \rightarrow A \\ A \rightarrow B \end{cases} \xrightarrow[\text{Simplify R.Hs}]{\text{Step-1,}} \begin{cases} AB \rightarrow C \\ B \rightarrow A \\ A \rightarrow B \end{cases}$$

Step-2
Eliminate Extraneous attribute from LHS

After eliminating "A" w.r.t. B

$$\begin{cases} B \rightarrow C \\ B \rightarrow A \\ A \rightarrow B \end{cases} \xrightarrow{\text{Step-3}} \text{Eliminate Redundant FDs}$$

No redundand FD
∴ FD set after eliminating all redundant FD $= \begin{cases} B \rightarrow C \\ B \rightarrow A \\ A \rightarrow B \end{cases}$

Step-2
Eliminate Extraneous attribute from LHS

after eliminating "B" w.r.t. A

$$\begin{cases} A \rightarrow C \\ B \rightarrow A \\ A \rightarrow B \end{cases} \xrightarrow{\text{Step-3}} \text{Eliminate Redundant FDs}$$

No Redundant RD
∴ FD set after eliminating all redundant FDs $= \begin{cases} A \rightarrow C \\ B \rightarrow A \\ A \rightarrow B \end{cases}$

#e.g., Consider the FD set

$F = \{AB \rightarrow C,\ B \rightarrow A,\ A \rightarrow B\}$

$F: \{AB \rightarrow C,\ B \rightarrow A,\ A \rightarrow B\}$ — Step-1, Simplify R.Hs. → $\{AB \rightarrow C,\ B \rightarrow A,\ A \rightarrow B\}$

Step-2: Eliminate Extraneous attribute from LHS

After eliminating "A" w.r.t. B: $\{B \rightarrow C,\ B \rightarrow A,\ A \rightarrow B\}$

Step-3: Eliminate Redundant FDs → $\{B \rightarrow C,\ B \rightarrow A,\ A \rightarrow B\}$

Step-4: Apply Union

$Fm_1 = \{B \rightarrow AC,\ A \rightarrow B\}$

it is minimal Cover of F

Step-2: Eliminate Extraneous attribute from LHS

after eliminating "B" w.r.t. A: $\{A \rightarrow C,\ B \rightarrow A,\ A \rightarrow B\}$

Step-3: Eliminate Redundant FDs → $\{A \rightarrow C,\ B \rightarrow A,\ A \rightarrow B\}$

Step-4: Apply Union

$\{A \rightarrow BC,\ B \rightarrow A\} = Fm_2$

Minimal Cover of F.



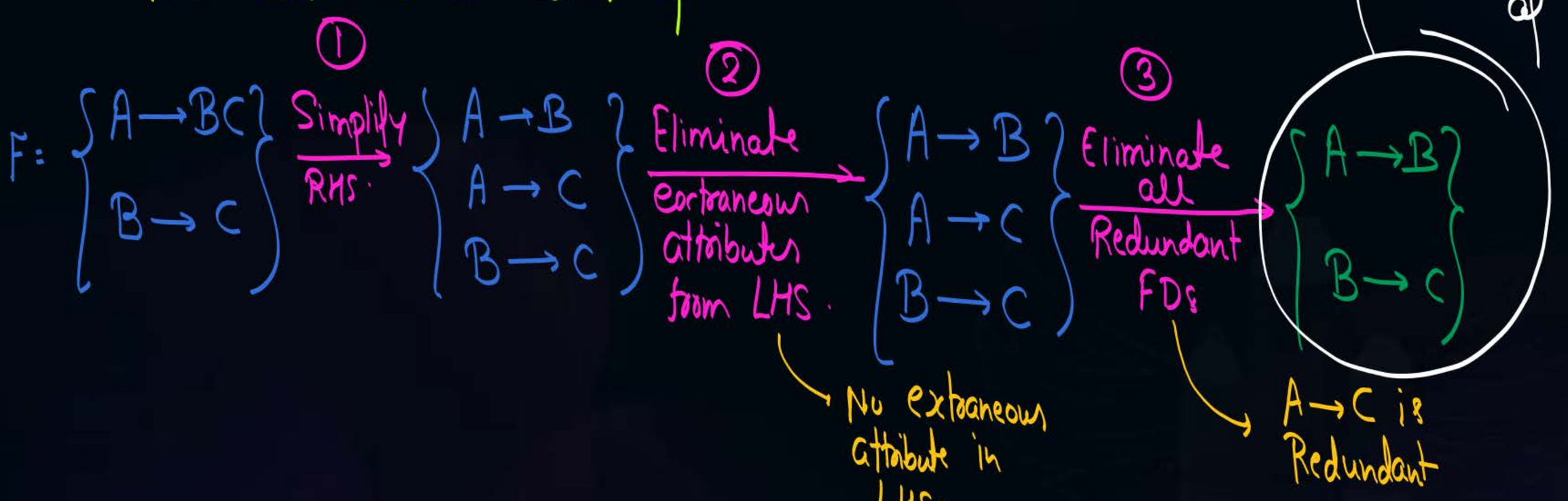
Topic : NOTE

- Minimal cover of FD set F need not be unique, but all minimal cover are logically equivalent.
  - i.e., If Fm_1 & Fm_2 are two minimal Covers of FD set F, then $Fm_1 \equiv Fm_2$
- Minimal Covers are also known as irreducible set of functional dependencies
  - irreducible → (Can not be reduced any further)

#e.g., Consider the FD set

$F = \{A \rightarrow BC,\ B \rightarrow C\}$

Find all minimal Covers of F.



#e.g., Consider the following FD set

F ={AC →G

D→EG

BC→D

CG→BD

ACD→B

CE→AG

}

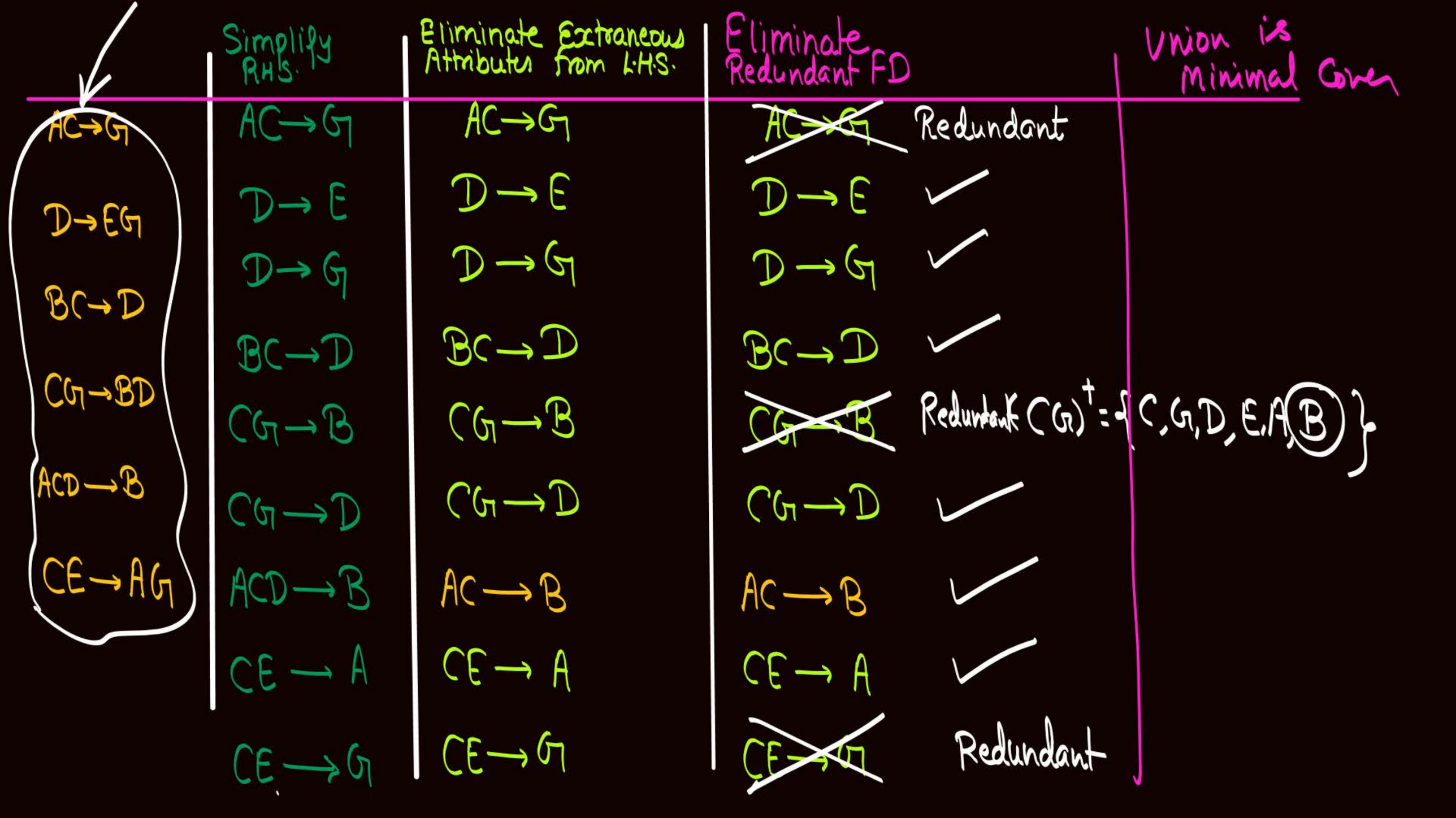
Find minimal cover of F.

	Simplify RHS.	Eliminate Extraneous Attributes from L.H.S.
$AC \to G$	$AC \to G$	$AC \to G$
$D \to EG$	$D \to E$	$D \to E$
	$D \to G$	$D \to G$
$BC \to D$	$BC \to D$	$BC \to D$
$CG \to BD$	$CG \to B$	$CG \to B$
$ACD \to B$	$CG \to D$	$CG \to D$
$CE \to AG$	$ACD \to B$	
	$CE \to A$	
	$CE \to G$	

$(AC)^+ = \{A, C, G, D, B, \ldots\}$ — D is extraneous w.r.t. (AC)

$(AD)^+ = \{A, D, E, G\}$ — $C \notin (AD)^+ \therefore C$ is not extraneous w.r.t (AD)

$(CD)^+ = \{C, D, E, G, B, A, \ldots\}$ — A is extraneous w.r.t (CD)


AC→G
D→EG
BC→D
CG→BD
ACD→B
CE→AG
Simplify RHS.
AC→G
D→E
D→G
BC→D
CG→B
CG→D
ACD→B
CE→A
CE→G
Eliminate Extraneous Attributes from L.H.S.
AC→G
D→E
D→G
BC→D
CG→B
CG→D
AC→B
CE→A
CE→G
Eliminate Redundant FD
AC→G Redundant
D→E ✓
D→G ✓
BC→D ✓
CG→B Redundant (CG)+ = {C, G, D, E, A, B}
CG→D ✓
AC→B ✓
CE→A ✓
CE→G Redundant
Union is Minimal Cover

	Simplify RHS.	Eliminate Extraneous Attributes from L.H.S.	Eliminate Redundant FD	
$AC \to G$	$AC \to G$	$AC \to G$	~~$AC \to G$~~	Redundant
$D \to EG$	$D \to E$	$D \to E$	$D \to E$	✓
	$D \to G$	$D \to G$	$D \to G$	✓
$BC \to D$	$BC \to D$	$BC \to D$	$BC \to D$	✓
$CG \to BD$	$CG \to B$	$CG \to B$	~~$CG \to D$~~	Redundant
$ACD \to B$	$CG \to D$	$CG \to D$	$CG \to B$	✓
$CE \to AG$	$ACD \to B$	$AC \to B$	$AC \to B$	✓
	$CE \to A$	$CE \to A$	$CE \to A$	✓
	$CE \to G$	$CE \to G$	~~$CE \to G$~~	Redundant

Union is Minimal Cover

$(CG)^+ = \{C, G, B, D, \ldots\}$

	Simplify RHS.	Eliminate Extraneous Attributes from L.H.S.	Eliminate Redundant FD		Union is Minimal Cover
AC→G	AC→G	AC→G	AC→G	✓	
D→EG	D→E	D→E	D→E	✓	
	D→G	D→G	D→G	✓	
BC→D	BC→D	BC→D	BC→D	✓	
CG→BD	CG→B	CG→B	~~CG→B~~	Redundant	
ACD→B	CG→D	CG→D	CG→D	✓	
CE→AG	ACD→B	CD→B	CD→B	✓	
	CE→A	CE→A	CE→A	✓	
	CE→G	CE→G	~~CE→G~~	Redundant	

	Simplify RHS.	Eliminate Extraneous Attributes from L.H.S.	Eliminate Redundant FD		Union is minimal Cover
$AC \to G$	$AC \to G$	$AC \to G$	$AC \to G$	✓	
$D \to EG$	$D \to E$	$D \to E$	$D \to E$	✓	
	$D \to G$	$D \to G$	$D \to G$	✓	
$BC \to D$	$BC \to D$	$BC \to D$	$BC \to D$	✓	
$CG \to BD$	$CG \to B$	$CG \to B$	~~$CG \to D$~~	Redundant	
$ACD \to B$	$CG \to D$	$CG \to D$	$CG \to B$	✓	
$CE \to AG$	$ACD \to B$	$CD \to B$	~~$CD \to B$~~	Redundant	
	$CE \to A$	$CE \to A$	$CE \to A$	✓	
	$CE \to G$	$CE \to G$	~~$CE \to G$~~	Redundant	

H.W.

#e.g., Consider the following FD set

$F = \{A \rightarrow BC$

$CD \rightarrow E$

$E \rightarrow C$

$D \rightarrow AEH$

$ABH \rightarrow BD$

$DH \rightarrow BC$

$\}$

Find minimal cover of F.



2 mins Summary

Topic Minimal Cover

Topic [redacted]

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THANK - YOU